Shivesh Kumar

Curriculum Vitae

"Wir müssen wissen – wir werden wissen" - David Hilbert

Vision & Research Interests

Vision — Next generation versatile automation with highly dynamic robots

Research Interests — geometry, kinematics, dynamics, optimal control theory, biomechanics, variable fidelity modeling, variable stiffness mechanisms, physical/athletic intelligence

Academic Degrees & Education

2025 — **SUHF Diploma in Teaching and Learning in Higher Education**, *Chalmers University of Technology*, Gothenburg, Sweden.
The Association of Swedish Higher Education (SUHF)

2019 — **Doctorate in Natural Sciences (Dr. rer. nat.)**, *Faculty of Mathematics & Computer Science, University of Bremen*, Bremen, Germany, *Distinction: Summa Cum Laude*.
Specialized in Computer Science

2013–2015 — **Masters of Science in Control Engineering, Robotics and Applied Informatics**, *Ecole Centrale de Nantes*, Nantes, France, *GPA – 15.7/20 (3 rd Rank in M1, 5 th Rank in M2)*.
Specialized in Advanced Robotics

2009–2013 — **Bachelor of Technology in Mechanical Engineering**, *National Institute of Technology Karnataka*, Surathkal, India, *GPA – 8.59/10 (Top 15% of the Batch)*.

2009 — **All India Senior School Certificate Examination**, *Central Board of Secondary Education*, Delhi, India, *Aggregate Marks: 88% (1 st Rank in School)*.

2007 — **All India School Certificate Examination**, *Central Board of Secondary Education*, Delhi, India, *Aggregate Marks: 89.2% (2 nd Rank in School)*.

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Date of Birth: 28.03.1992, Nationality: Indian, Permanent Resident: EU

Dissertation & Theses

Doctoral Dissertation

Title *Modular and Analytical Methods for Solving Kinematics and Dynamics of Series-Parallel Hybrid Robots*

Supervisors Prof. Dr. Frank Kirchner (Universität Bremen, Germany) & Prof. Dr. Andreas Müller (Johannes Kepler Universität Linz, Austria)

Acquired Skills Screw Theory, Lie Group Theory, Computational Algebraic Geometry, Kinematics and Dynamics, Equations of Motion, Holonomic Constraints, Complex Mechanisms.

Master Thesis

Title *Contribution to Modeling of Human Walking Gait Cycle over Stride based on Robotics for Pedestrian Navigation Solution*

Supervisors Prof. Valerie Renaudin (Université Gustave Eiffel) & Prof. Yannick Aoustin (Université de Nantes), Prof. Eric Le-Carpentier (Ecole Centrale de Nantes, France)

Acquired Skills Biomechanics, Gait Analysis, Optimal Control, Sequential Quadratic Programming, Pedestrian Navigation, Dynamic Modeling and Parameter Estimation of Humans.

Bachelors Thesis

Title *All-Terrain Mobile Robot for Extra-Terrestrial Applications*

Supervisors Prof. Dr. K V Gangadharan (National Institute of Technology Karnataka, India)

Acquired Skills Multi–Body Dynamics Simulation, RecurDyn, Co-simulation, Mechatronics System Design of Shrimp Rover, Embedded Systems Programming, CAD Modeling

Experience

Professional

October 2023 – Present **Assistant Professor (Tenure Track)**, *Department of Mechanics and Maritime Sciences, Chalmers University of Technology*, Gothenburg, Sweden.

Topic: Dynamics and Control of Mechanical Systems, Division: Dynamics
My primary responsibilities for this position include:
- Develop research within the relevant field in cooperation with academy and industry
- Participate in and develop teaching on bachelor and master's levels within the solid mechanics field
- Attract funding from industrial partners and research agencies (successful with attracting approx. 22.4 Million SEK until now)
- Contribute to research centers at Chalmers, e.g., the six Areas of Advance .
- Take an active part in contributing to a well-functioning academic environment and collegial activities at the department
- Contribute to fulfilling the department's goals on outreach activities together with current and future partners, within Sweden and internationally

June 2024– February 2025 **Contracted Senior Researcher (10%)**, *Robotics Innovation Center, German Research Center for Artificial Intelligence (DFKI GmbH)*, Bremen, Germany.
- Advisory Relationship to Team Mechanics & Control and Underactuated Robotics Lab
- PhD Supervision: Ibrahim Tijjani (graduated September 2024), Bingbin Yu (March 2025), Shubham Vyas, Felix Wiebe
- Transfer of project leadership in VeryHuman and M-RoCK SAB projects
- Collaboration on Quantum Optimal Control

July 2020–May 2024 **Senior Researcher**, *Robotics Innovation Center, German Research Center for Artificial Intelligence (DFKI GmbH)*, Bremen, Germany.

Detailed achievements:
- Project Acquisition:
  - Successful: VeryHuman (BMBF), M-RoCK (BMBF), PACOMA (ESA), AAPLE (DLR)
  - In pipe-line: SFB ARISE A5 (DFG)
- Project Leader of the VeryHuman project (until October 2023)
  - Aim: Learning and verifying complex behaviours for humanoid robots
  - learning and control framework for a complex series-parallel humanoid robot
- Deputy Project Leader of the M-RoCK project (until October 2023)
  - Aim: Modelling of human-machine interaction for improvement of robot behaviour
  - Integration of implicit and explicit human feedback for improving optimal control
- Lab Supervisor of the Underactuated Robotics Lab at DFKI-RIC (until October 2023)
- Lead Developer of Hybrid Robot Dynamics (HyRoDyn) software framework
  - kinematics and dynamics solver for highly complex series-parallel hybrid robotic systems
  - at the heart of various robotic systems at DFKI-RIC (e.g. RH5 humanoid, Recupera full body exoskeleton, Recupera wheelchair exoskeleton)
- Lead Editor of a book based on my PhD work titled *"Biologically Inspired Series-Parallel Hybrid Robots: Design, Analysis and Control"* co-edited with my supervisors Prof. Andreas Müller (JKU Linz) and Prof. Frank Kirchner (Uni Bremen) to be published by Elsevier, USA in 2023.
- External scientific collaboration with Prof. Andreas Mueller (JKU, Linz), Dr. Abhinandan Jain (NASA JPL, Pasadena), Dr. Abhilash Nayak (LS2N, Nantes), Prof. Damien Chabalat (LS2N, Nantes), Prof. Suril Shah (IIT Jodhpur), Dr. Olivier Stasse (LAAS-CNRS, Toulouse)

April 2019–October 2023 **Team Leader: Mechanics & Control**, *Robotics Innovation Center, German Research Center for Artificial Intelligence (DFKI GmbH)*, Bremen, Germany.

The Advanced AI team Mechanics & Control which is composed of 8 PhD candidates and 2 Post Doctoral researcher studies the fundamentals of mechanics and motion control for robotics applications by taking inspiration from the mathematical fields of analytical mechanics, differential geometry and algebraic geometry. The team is working on various cutting edge research topics for e.g. optimal control of humaoid robot using differential dynamic programming, modeling of continuum manipulator, variable stiffness mechanisms, fluid-structure interaction, motion planning in discrete trajectory space etc.

In my role as the team leader, I am responsible for:
- formulation of team vision and goals
- project acquisition for the team (total budget: 919,639 Euros in 2021)
- scientific supervision of the members and quality control
- organizing team's journal club
- transfer of knowledge and software to other application oriented teams at DFKI-RIC.

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October 2015–June 2020 **Researcher**, *Robotics Innovation Center, German Research Center for Artificial Intelligence (DFKI GmbH)*, Bremen, Germany.

Worked in Robot Control Team and contributed to various R & D (Recupera-Reha, D-RoCK, Q-RoCK) and industrial (ILAADR) projects.

- *Q-RoCK: AI-based Qualification of Deliberative Behaviour for a Robotic Construction Kit* (01.08.2018 to 31.07.2021, funded by BMBF, monitored by Scientific Advisory Board)
  - Basic research on robot's capabilities that can be derived from its hardware specifications
  - Supervised research on combinatronics of a discrete trajectory space for motion planning
  - Model simplification for dynamic control of series–parallel hybrid robots
- *TransFIT: Flexible Interaction for infrastructures establishment by means of teleoperation and direct collaboration; transfer into industry 4.0* (01.07.2017 to 30.06.2021, funded by BMBF, monitored by DLR)
  - Teleoperation with force feedback
  - Architect of mid-level control software of the humanoid and exoskeleton
  - Supervised research on variable stiffness mechanisms
- *Recupera-Reha: Full-body exoskeleton for upper body robotic assistance* (01.09.2014 to 31.12.2017, funded by BMBF, monitored by DLR)
  - Kinematic and dynamic modeling of Recupera exoskeleton for rehabilitation application
  - Implementation of various therapy modes e.g. teach & replay, mirror therapy, gravity compensation and conducted successful trials with stroke patients.
  - Software architect for the mid-level control of the Recupera-Reha exoskeleton.
- *D-RoCK: Models, methods and tools for the model based software development of robots* (01.06.2015 to 31.05.2018, funded by BMBF, monitored by Scientific Advisory Board)
  - Modular and analytical approach for kinematic and dynamic modeling for complex robots
  - First software implementation of HyRoDyn.
- *ILAADR: Internal Logistics with Automated Autonomous Delivery and Replenishment* (01.01.2016 to 31.12.2016 , funded by EIT Digital)
  - Low level drivers for UR10 robot and Robotiq two finger gripper
  - Computer vision software for manipulation with HALCON library
  - Performed tests at industrial site of FIAT Research Center in Torino, Italy.

In addition to my involvement in the above project work, I also took the following roles:

- Represented associated projects at various public events e.g. Hannover Messe, CEBIT, RIC Tag der Offenen Tür (Open Day).
- System responsible of UR10 robot from Universal Robots and Software responsible for software tools HyRoDyn and CAD2SIM.
- Organizer of study group on "Differential Geometry".

Teaching

March 2025–May 2025 **TRA455: Athletic Intelligence in Robotics**, *Chalmers University of Technology*, Gothenburg, Sweden.

Masters level Tracks course (7.5 Credits)

November 2024–December 2024 **TME141: Structural Dynamics**, *Chalmers University of Technology*, Gothenburg, Sweden.

Masters level course (7.5 Credits) focussing on computational aspects of linear structural dynamics. 15 Lectures, Computer Assignment Sessions, Consultancy Hours

June 2024 **MBCAIR: Model Based Control for Athletic Intelligence in Robots**, *Ecole Centrale de Nantes*, Nantes, France.

Visiting professor (Masters Level).

April 2024–August 2024 **MRCA: Modern Robot Control Architectures**, *University of Bremen*, Bremen, Germany.
Visiting professor to teach part of the MRCA course (Bachelors+Masters Level):

March 2024–May 2024 **Dynamics of Multibody Systems**, *Chalmers University of Technology*, Gothenburg, Sweden.
PhD level course taught with Prof. Håkan Johansson, contributed to lectures, exercise and tutorials on the minimal coordinate formulation of the multi-body dynamics:

November 2023–December 2023 **TME141: Structural Dynamics**, *Chalmers University of Technology*, Gothenburg, Sweden.
11 Exercise Sessions on various topics in Structural Dynamics, assisted Prof. Thomas Abrahmsson (retired) in the computer assignment and consultancy sessions with students.

April 2023–August 2023 **MRCA: Modern Robot Control Architectures**, *University of Bremen*, Bremen, Germany.

September 2022–March 2022 **Propaedeutic C/C++**, *University of Bremen*, Bremen, Germany.
Hands-on Tutorials on C/C++ programming

April 2022–August 2022 **MRCA: Modern Robot Control Architectures**, *University of Bremen*, Bremen, Germany.
Contributed to restructuring the course and modernizing it.

April 2022–August 2022 **HCIR: Human Centered Interaction in Robotics**, *University of Bremen*, Bremen, Germany.

April 2021–August 2021 **MRCA: Modern Robot Control Architectures**, *University of Bremen*, Bremen, Germany.

Academic Exchange Stays

August 2019–September 2019 **Visiting Researcher with Dr. Abhinandan Jain**, *Robotics Modeling and Simulation Group, NASA Jet Propulsion Laboratory*, Pasadena, USA.
- Attended Dynamics Algorithms for Real Time Simulation (DARTS) Lab Workshop to gain a first hand experience with several multi-mission space system simulation tools (e.g. DARTS, DSHELL, DSENDS, ROAMS, PyCraft etc.) for the closed-loop development and testing of flight algorithms and software, acquired DARTS license for use in DFKI-RIC.
- Learnt about spatial operator algebra (SOA) and constraint embedding approaches for modeling of multi-body systems
- Validation study of HyRoDyn model of a complex series-parallel hybrid humanoid leg in DARTS software

June 2019 **Visiting Researcher with Prof. Dr. Andreas Müller**, *Institute of Robotics, Johannes Kepler University*, Linz, Austria.
- PhD thesis write-up and corrections
- Invited talk at Institute of Robotics at JKU Linz
- Preparation of tutorial on Lie Group Modeling of Robotic Systems

September 2017 **Visiting Researcher with Prof. Dr. Andreas Müller**, *Institute of Robotics, Johannes Kepler University*, Linz, Austria.
- Kinematic analysis of 2SPRR+1U parallel mechanism using computational algebraic geometry
- Modular and analytical methods for solving kinematics and dynamics of series-parallel hybrid robots
- Singularity School 2017

July 2014–August 2014 **Summer Intern with Dr. Nicolas Mansard**, *Humanoid Robotics Group (GEPETTO), Laboratory for Architecture of Autonomous Systems (LAAS-CNRS)*, Toulouse, France.
Developed RBDLPY (Python interface for Rigid Body Dynamics Library using Boost Python), Simulation of rigid body contacts in multi-body systems using RBDLPY

May 2012–July 2012 **Summer Intern with Mr. Simon Althoff**, *Institute of Mechatronics & Dynamics, University of Paderborn*, Paderborn, Germany.
Modal Analysis and Modal Reduction of any system using MATLAB and Ansys.

May 2011–July 2011 **Summer Intern with Prof. Subir Kumar Saha**, *Mechatronics Lab, Indian Institute of Technology Delhi*, New Delhi, India.
Robot Modeling & Dynamic Simulation using RecurDyn & Autodesk Inventor, Development of an Automatic DH Parameter Extraction Algorithm and its implementation as an addon in Autodesk Inventor

May 2010–July 2010 **Summer Intern**, *ThinkLABS, Indian Institute of Technology Bombay*, India.
Training in the field of Embedded Systems and Robotic Manipulator design, Project – Design and Development of a Card Dealer Robotic Arm

Awards & Recognitions

2025 Future Research Leader Grant from Swedish Foundation for Strategic Research

2025 Associate Editor, IEEE Robotics & Automation Letters journal

2025 Associate Editor, IEEE/RSJ IROS 2025, Hangzhou, China

2024 Recognized as a *New Generation Star* in Robotics Research at the IEEE/RSJ IROS 2024, Abu Dhabi, UAE, sponsored by NOKOV Motion Capture

2024 Associate Editor, IEEE/RSJ IROS 2024, Abu Dhabi, UAE

2024 2024 EuroGNC Best Paper Award

2024 2023 Outstanding Paper Award for Young Scientists by Committee on Space Research, French National Space Agency (COSPAR-CNES)

2024 Invited Member of the International Scientific Committee of the International Conference Series on Climbing and Walking Robots and the Support Technologies for Mobile Machines (CLAWAR)

2022 Finalist of IROS 2022 Best Entertainment and Amusement Paper Award

2022 Associate Co-Chair of IEEE Robotics and Automation Society (RAS) Technical Committee on *Model Based Optimization for Robotics*

2017 2nd prize in Best Oral Presentations at Advances in Robotics Conference (AIR 2017) sponsored by Springer Nature

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2013–2015	Erasmus Mundus HERITAGE scholarship to pursue my Masters studies in Advanced Robotics at Ecole Centrale de Nantes
2013–2015	Erasmus Mundus India4EU scholarship to pursue my Masters studies in Mechatronics at Politechnico di Torino (declined against HERITAGE offer)
2012	Conference grant from Function Dynamics India Pvt. Ltd. for presenting my paper in Asian Conference on Multi-body Dynamics (ACMD) 2012 in Shanghai, China
2012	DAAD-IAESTE scholarship for research stay at Institute of Mechatronics and Dynamics, University of Paderborn
2009–2013	CBSE Central Sector Scholarship for pursuing B. Tech. degree in Mechanical Engineering at NITK Surathkal, India

Computer Skills & Competences

Programming	C, C++, Ruby, Python
Scientific Computing	MATLAB/Simulink, Maple, Mathematica
CAD	Autodesk Inventor, Solidworks, Catia
CAE	RecurDyn, ADAMS MSC, ANSYS
Standard Robots	Universal Robots UR5 and UR10, Staübli RX90, PUMA 560, Baxter (Rethink Robotics)
DFKI-RIC Robots	Recupera Wheelchair Exoskeleton, Recupera Full Body Exoskeleton, RH5 Humanoid
Embedded Systems	Atmega16, Atmega2560
Version Control	Git, SVN
Office	LaTeX, Libreoffice, Microsoft Office
Operating Systems	Ubuntu (Linux), Microsoft Windows, Robot Construction Kit (RoCK), Robot Operating System (ROS)

Invited Talks or Guest Lectures

14 July 2025	Introduction to Robotics & Dynamics in IEEE-RAS Summer School on Optimization for Robotics, Patras, Greece.
26 March 2024	RealAIGym: Education and Research Platform for Athletic Intelligence in Workshop on Learning and Control of Dynamic Robots, KTH Royal Institute of Technology, Stockholm, Sweden.
14 March 2024	Overview of My Research on Dynamics and Control of Robotic Systems, ABB Corporate Research Center, Västerås, Sweden.
25 September 2023	Athletic Intelligence in Robotics, PIP Summer School on Machine Learning and its Applications, University of Bremen, Germany.
12 April 2023	RealAIGym: Education and Research Platform for Athletic Intelligence in Workshop on Learning and Control of Dynamic Robots, Fraunhofer IML, Dortmund, Germany.

16 December 2022 Some Mathematical Pitfalls in Robotics in the course "Prozessinformatik" at Institut für Robotik und Prozessinformatik, TU Braunschweig, Germany.

15 March 2021 Lie group based modelling of robot kinematics & dynamics and an optimization based view on robotics at Space environment stability and resilience – Stardust-R Network Training School III, Iasi, Romania (held virtually).

14 December 2020 An Optimization Based View on Robot Dynamics in the course "Prozessinformatik" at Institut für Robotik und Prozessinformatik, TU Braunschweig, Germany.

13 December 2019 Overview of Robotics Activities at DFKI-RIC and A Software Architecture for Solving Kinematics and Dynamics of Series-Parallel Hybrid Robots at Amrita School of Engineering (ASE), Bengaluru, India.

05 September 2019 Overview of Robotics Activities at DFKI-RIC and A Software Architecture for Solving Kinematics and Dynamics of Series-Parallel Hybrid Robots at NASA Jet Propulsion Laboratory, Pasadena, United States

13 July 2019 A modular approach for kinematic and dynamic modeling of complex robotic systems using algebraic geometry, at the mini-symposium on Algebraic Geometry for Kinematics and Dynamics in Robotics at SIAM Conference on Applied Algebraic Geometry, Bern, Switzerland (co-authored with Prof. Andreas Mueller).

26 June 2019 A Modular Software Workbench for Kinematic and Dynamic Modeling of Complex Series-Parallel Hybrid Robots at Institute of Robotics (ROBIN), Johannes Kepler University, Linz, Austria.

11 September 2018 Modular and Distributable Approach towards Kinematic and Dynamic Modeling of Series-Parallel Hybrid robots at Institut für Robotik und Prozessinformatik, TU Braunschweig, Germany.

06 December 2017 Design, analysis and control of a novel almost spherical mechanism: Active Ankle, In Kinematics, Dynamics and Mechatronics in Motion Technology - Seminar RWTH Aachen, at Institut für Getriebetechnik, Maschinendynamik und Robotik, RWTH Aachen, Germany.

Services to the Scientific Community/Academic Citizenship

Chair

1. Sessions on Robust and Adaptive Control I & II, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2024), Abu Dhabi, UAE.
2. Session on Constraint Satisfaction and Optimization, 2024 International Joint Conference on Artificial Intelligence, Jeju Island, Republic of Korea.
3. Session on Robotics V, 2022 IEEE 61st Conference on Decision and Control (CDC), Cancún, Mexico (invited, but declined).
4. Session on Humanoid Systems, IEEE International Conference on Robotics and Automation (ICRA 2022), Philadelphia, USA.
5. Session on Dynamics, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2019), Macau, China
6. Session on AI in Robotics – Applications, 52nd ISR : International Symposium on Robotics (ISR 2020), Munich, Germany (held virtually)
7. Panelist on Role of AI and Robotics in Digitalization – German–India Business Summit 2020, Bremen, Germany (held virtually)

Workshops, Tutorials, Competitions

1. Competition titled *3rd AI Olympics With RealAIGym: Towards Global Swing-up Policies* at IEEE ICRA 2025, Atlanta, USA, co–organized with Prof. Kirchner (DFKI Bremen) and Prof. Jan Peters (DFKI Darmstadt).
2. Competition titled *2nd AI Olympics With RealAIGym: Is AI Ready for Athletic Intelligence in the Real World?* at IEEE IROS 2024, Abu Dhabi, UAE, co–organized with Prof. Kirchner (DFKI Bremen) and Prof. Jan Peters (DFKI Darmstadt).
3. Workshop titled *Loco-Manipulation: Algorithms, Challenges & Applications* at IEEE ICRA 2024, Yokohama, Japan, co–organized with Oliver Urbann (Fraunhofer IML), Julian Esser (Fraunhofer IML), Ioannis Havoutis (Oxford University), Gabriel Margolis (MIT), Carlos Mastalli (Heriot-Watt University), Claudio Semini (IIT), Olivier Stasse (LAAS-CNRS).
4. Workshop titled *Co-design in Robotics: Theory, Practice, and Challenges* at IEEE ICRA 2024, Yokohama, Japan, co–organized with Cynthia Sung (GRASP Lab, UPenn), Darwin Lau (Chinese University of Hong Kong), Hannah Stuart (UC Berkeley), Pauline Pounds (University of Queensland), Patrick M. Wensing (Uni Notre Dame).
5. Competition titled *1st AI Olympics With RealAIGym: Is AI Ready for Athletic Intelligence in the Real World?* at IJCAI 2023, Macao, co–organized with Prof. Jan Peters (TU Darmstadt)
6. Tutorial titled *Getting the Right Model of your Robot in an Easy way using Modern Geometric Methods* at ASME IDETC 2022, St. Louis, USA, co–organized with Prof. Andreas Müller (JKU Linz)
7. Tutorial titled *Lie Group Modeling of Robot Kinematics and Dynamics – A Hands-On Introduction* at ASME IDETC 2019, Anaheim, USA, co–organized with Prof. Andreas Müller (JKU Linz)

Master Thesis Supervision
1. C. Stoeffler, Conceptual Design of a Variable Stiffness Mechanism in a Humanoid Ankle using Parallel Redundant Actuation, University of Liége, June 2018.
2. Tim von Oehsen, Systematische Analyse maschineller Lernverfahren zur Lösung des Problems der inversen Kinematik, University of Bremen, June 2019.
3. Durgesh H. Salunkhe, Optimal Kinematic Design of Parallel Mechanism for Otological Surgery, Ecole Centrale de Nantes, August 2020.
4. Julian Esser, Highly-Dynamic Movements of a Humanoid Robot Using Whole-Body Trajectory Optimization, University of Duisburg–Essen, September 2020.
5. Rohit Kumar, A hybrid numerical and analytical approach towards resolving loop closure constraints in rigid body dynamics, University of Genoa, August 2021.
6. Mareike Foerster, Perception thresholds and discrimination: Force feedback in an exoskeleton-based teleoperation of a humanoid robot, City University of Applied Sciences, Bremen, May 2022.
7. Mahdi Javadi, Mechatronic Design and Control of an Underactuated Brachiation Robot, TU Kaiserslautern, Germany, July 2022.
8. Federico Girlanda, Robust Co-design of Underactuated Mechanical Systems, University of Padova, Italy, April 2023.
9. Grama Srinivas Shourie, Mechatronic Design and Control of an Underactuated Three-link Brachiation Robot, Ecole Centrale de Nantes, France, August 2023.
10. Maximilian Albracht, Control of Dynamic Parkour Motions for a Hopping Leg on a Broomstick, FH Aachen University of Applied Sciences, Germany, August 2023.
11. Chi Zhang, Reinforcement Learning Based Control of an Underactuated Double Pendulum System, Technical University of Munich, Germany, November 2023.
12. Vinayvivian Pedrick Rodrigues, Kinetostatic Analysis of a 6-$\underline{R}$US Parallel Continuum Robot, RWTH Aachen, Germany, December 2023.

Consultant: Exhibition on History of Robotics, Futurium, Berlin, Germany

Journal Reviewer: International Journal of Robotics Research (IJRR), IEEE Robotics and Automation Letter, IEEE Robotics and Automation Magazine, IEEE Transactions on Mechatronics, Elsevier Robotics and Autonomous Systems, Springer Journal of Intelligent and Robotic Systems, ASME Journal of Mechanical Design (JMD), ASME Computational Nonlinear Dynamics (CND), MDPI Sensors, MDPI Applied Sciences, MDPI Robotics, IEEE Access, Robotica, ASME Journal of Mechanisms and Robotics

Conference Reviewer: International Conference on Robotics and Automation (ICRA), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Robotics Science and Systems (RSS), IEEE International Conference on Humanoid Roobtics (ICHR), Asian Mechanism and Machine Science (MMS), IEEE ASME IDETC Mechanisms and Robotics Conference, Advances in Robot Kinematics (ARK), BioRob, EuroHaptics

Languages

Hindi **Mothertongue**

English **Mothertongue**

German **Basic** *A2*

French **Basic** *A2*

Interests

- Mathematics
- Traveling
- Jogging
- Reading
- Cooking
- Cycling

List of Publications

Book

Shivesh Kumar, Andreas Mueller, Frank Kirchner. Biologically Inspired Series-Parallel Hybrid Robots: Design, Analysis and Control. Elsevier Academic Press, 2025, `https://doi.org/10.1016/C2020-0-02362-5`

Journal

Maximilian Albracht, Shivesh Kumar, Shubham Vyas, and Frank Kirchner. Model predictive parkour control of a monoped hopper in dynamically changing environments. *IEEE Robotics and Automation Letters*, 9(10):8507–8514, 2024

Gabriele Fadini, Shivesh Kumar, Rohit Kumar, Thomas Flayols, Andrea Del Prete, Justin Carpentier, and Philippe Souères. Co-designing versatile quadruped robots for dynamic and energy-efficient motions. *Robotica*, 42(6):2004–2025, 2024

Felix Wiebe, Shivesh Kumar, Lasse J. Shala, Shubham Vyas, Mahdi Javadi, and Frank Kirchner. Open source dual-purpose acrobot and pendubot platform: Benchmarking control algorithms for underactuated robotics. *IEEE Robotics & Automation Magazine*, 31(2):113–124, 2024

Mahdi Javadi, Daniel Harnack, Paula Stocco, Shivesh Kumar, Shubham Vyas, Daniel Pizzutilo, and Frank Kirchner. Acromonk: A minimalist underactuated brachiating robot. *IEEE Robotics and Automation Letters*, 8(6):3637–3644, 2023

Andreas Müller, Shivesh Kumar, and Thomas Kordik. A recursive lie-group formulation for the second-order time derivatives of the inverse dynamics of parallel kinematic manipulators. *IEEE Robotics and Automation Letters*, 8(6):3804–3811, 2023

Mihaela Popescu, Dennis Mronga, Ivan Bergonzani, Shivesh Kumar, and Frank Kirchner. Experimental investigations into using motion capture state feedback for real-time control of a humanoid robot. *Sensors*, 22(24):9853, 2022

Shubham Vyas, Lasse Maywald, Shivesh Kumar, Marko Jankovic, Andreas Mueller, and Frank Kirchner. Post-capture detumble trajectory stabilization for robotic active debris removal. *Advances in Space Research*, 2022

Durgesh Haribhau Salunkhe, Guillaume Michel, Shivesh Kumar, Marcello Sanguineti, and Damien Chablat. An efficient combined local and global search strategy for optimization of parallel kinematic mechanisms with joint limits and collision constraints. *Mechanism and Machine Theory*, 173:104796, 2022

Felix Wiebe, Jonathan Babel, Shivesh Kumar, Shubham Vyas, Daniel Harnack, Melya Boukheddimi, Mihaela Popescu, and Frank Kirchner. Torque-limited simple pendulum: A toolkit for getting familiar with control algorithms in underactuated robotics. *Journal of Open Source Software*, 7(74):3884, 2022

Ibrahim Tijjani, Shivesh Kumar, and Melya Boukheddimi. A survey on design and control of lower extremity exoskeletons for bipedal walking. *Applied Sciences*, 12(5):2395, 2022

Andreas Müller and Shivesh Kumar. Closed-form time derivatives of the equations of motion of rigid body systems. *Multibody System Dynamics*, pages 1–17, 2021

Thomas M. Roehr, Daniel Harnack, Hendrik Wöhrle, Felix Wiebe, Moritz Schilling, Oscar Lima, Malte Langosz, Shivesh Kumar, Sirko Straube, and Frank Kirchner. A development cycle for automated self-exploration of robot behaviors. *AI Perspectives*, 3(1), 7 2021

Shivesh Kumar, Kai Alexander von Szadkowski, Andreas Mueller, and Frank Kirchner. An Analytical and Modular Software Workbench for Solving Kinematics and Dynamics of Series-Parallel Hybrid Robots. *Journal of Mechanisms and Robotics*, 12(2), apr 2020

Shivesh Kumar, Hendrik Wöhrle, José de Gea Fernández, Andreas Müller, and Frank Kirchner. A survey on modularity and distributivity in series-parallel hybrid robots. *Mechatronics*, 68:102367, jun 2020

Shivesh Kumar, Hendrik Wöhrle, Mathias Trampler, Marc Simnofske, Heiner Peters, Martin Mallwitz, Elsa Kirchner, and Frank Kirchner. Modular Design and Decentralized Control of the Recupera Exoskeleton for Stroke Rehabilitation. *Applied Sciences*, 9(4):626, feb 2019

Anirvan Dutta, Durgesh Haribhau Salunkhe, Shivesh Kumar, Arun Dayal Udai, and Suril V. Shah. Sensorless full body active compliance in a 6 DOF parallel manipulator. *Robotics and Computer-Integrated Manufacturing*, 59:278–290, oct 2019

S. Kumar, B. Bongardt, M. Simnofske, and F. Kirchner. Design and Kinematic Analysis of the Novel Almost Spherical Parallel Mechanism Active Ankle. *Journal of Intelligent and Robotic Systems: Theory and Applications*, 2018

Book Chapters

Shivesh Kumar, Andreas Müller, and Frank Kirchner. Chapter 1 - motivation. In Shivesh Kumar, Andreas Müller, and Frank Kirchner, editors, *Biologically Inspired Series-Parallel Hybrid Robots*, pages 3–15. Academic Press, 2025

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